

Cloud Computing:

- Cloud computing is the delivery of services like:
 - o Servers
 - o Storage
 - o Network
 - o Databases
 - o Monitoring tools
 - o AI
- These services are delivered over the internet, and work on the principle “Pay-As-You-Go”.
- Pay-As-You-Go, means that you will only pay for the services that you use.
- Cloud computing is defined as *renting the IT services* over the internet.
- In Cloud computing, the core services are:
 - o Compute (CPU + RAM)
 - o Storage
 - o Network

Cloud computing VS On-premises

On-premises

- ✓ Within your own premises (house, building, campus, within organization).
- ✓ You need to take care of:
 - o Spaces
 - o Electricity
 - o Cooling
 - o Security
- ✓ The major advantage of on-prem is that the data stays in-house. No 3rd party intervention.

Cloud computing

- ✓ Over the internet, in some-one else’s datacenter.
- ✓ Vendors like:
 - o Google
 - o Microsoft
 - o Amazon
 - o Redhat
 - o VMWare
 - o Rackspace
- ✓ These vendors are providing the cloud computing capabilities with a small fee.

Cloud computing features:

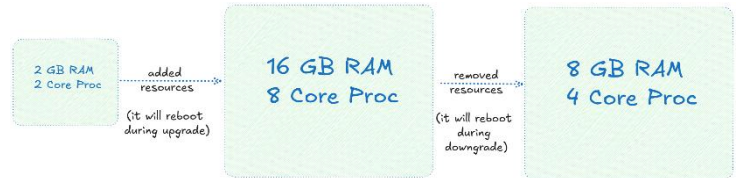
- On-demand
 - o The required service (compute, storage, network, database, etc.) must be available immediately when it is needed.
- Scalable
 - o It’s the ability to increase or decrease resources.



- Types of scalabilities:

○ *Vertical scaling*

- Scaling up/down
- Simple and faster.
- No changes in the architecture.
- It needs down time for upgrade and downgrade.
- Here, we add more resources to the existing VM.



○ *Horizontal scaling*

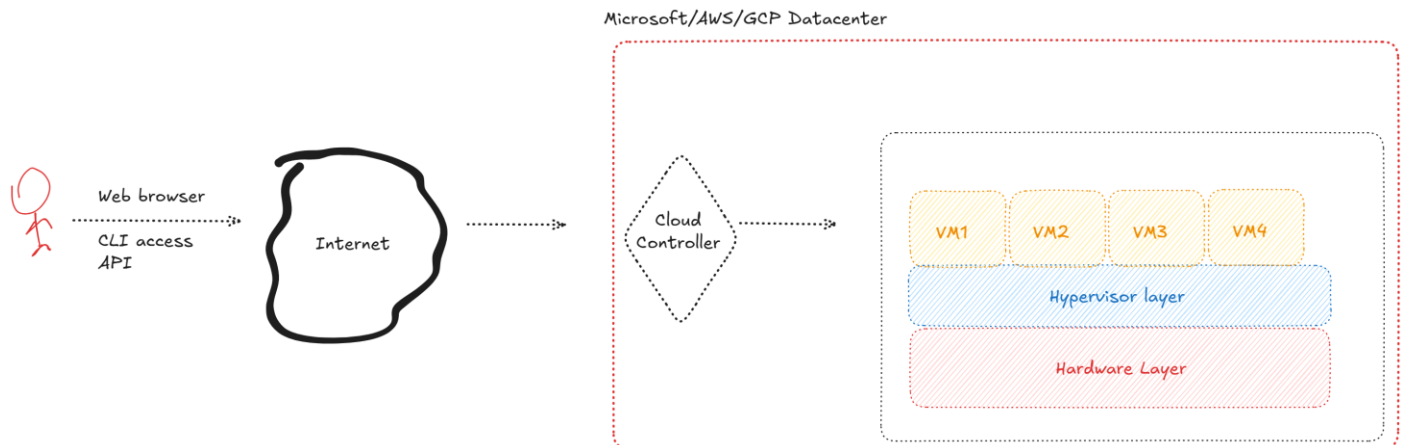
- Scaling out/In
- Adding more instances to handle the traffic.
- Scaling out → adding new instances
- Scaling in → removing instances
- No upper limit
- This provides the HA (high availability) and redundancy concepts.
- This is usually used in:
 - Amazon and Flipkart big billion sales.
 - IPL matches
 - Etc...
- It works using load balancer to manage the traffic.



- Unlimited computation and storage

- No limit on computation and storage.
- No limit on storing data on cloud, what ever you upload you need to pay for that.
 - Whenever you upload a file on cloud, it makes total of 3 copies in the backend.

Cloud architecture:



Cloud deployment models:

- *Public cloud*

- This cloud can be access by ANYONE over internet.
- This type of cloud has multi-tenancy.
- Example:
 - Microsoft Azure
 - Amazon web service (AWS)
 - Google Cloud Platform (GCP)
 - Redhat cloud
 - VMWare cloud
 - Digital Ocean
 - Rackspace
 - Oracle Cloud

- *Private cloud*
 - Private cloud has single-tenancy.
 - People from same organization can create their own resources under private cloud.
 - Examples:
 - Microsoft System Center Suite
 - Openstack (Linux-based) → Rackspace + NASA
- *Hybrid cloud*
 - Combination of Public and Private cloud.
- *Community cloud*
 - For specific communities
 - Example:
 - Gov cloud
 - Government cloud for govt. employees.
 - China cloud
 - Germany cloud
- *Inter-cloud models*
 - Combination of various cloud
 - AWS + Azure
 - Azure + GCP
 - AWS + GCP



-  Elastic Compute Cloud (EC2)
-  Elastic Kubernetes Service (EKS)
-  Lambda
-  Simple Storage Service (S3)
-  Elastic Block Store
-  Elastic File System
-  Virtual Private Cloud
-  Route 53
-  Elastic Load Balancing
-  Web Application Firewall
-  RDS
-  DynamoDB
-  Redshift
-  Elastic MapReduce
-  Kinesis
-  SageMaker
-  Glue
-  EventBridge
-  Simple Queuing Service
-  Simple Notification Service
-  CloudWatch
-  CloudFormation
-  IAM
-  KMS



-  Virtual Machine
-  Azure Kubernetes Service (AKS)
-  Azure Functions
-  Blob Storage
-  Managed Disk
-  File Storage
-  Virtual Network
-  DNS
-  Load Balancer
-  Web Application Firewall
-  SQL Database
-  Cosmos DB
-  Synapse Analytics
-  HDInsight
-  Streaming Analytics
-  Machine Learning
-  Data Factory
-  Event Grid
-  Storage Queues
-  Service Bus
-  Monitor
-  Resource Manager
-  Active Directory
-  Key Vault



-  Compute Engine
-  Google Kubernetes Engine (GKE)
-  Cloud Functions
-  Cloud Storage
-  Persistent Disk
-  File Store
-  Virtual Private Cloud
-  Cloud DNS
-  Cloud Load Balancing
-  Cloud Armor
-  Cloud SQL
-  Firebase Realtime Database
-  BigQuery
-  Dataproc
-  Dataflow
-  Vertex AI
-  Data Fusion
-  Eventarc
-  Pub/Sub
-  Firebase Cloud Messaging
-  Cloud Monitoring
-  Deployment Manager
-  Cloud Identity
-  Cloud KMS



-  Virtual Machine
-  Oracle Container Engine
-  OCI Functions
-  Object Storage
-  Persistent Volume
-  File Storage
-  Virtual Cloud Network
-  DNS
-  Load Balancer
-  Web Application Firewall
-  ATP
-  NoSQL Database
-  Autonomous Data Warehouse
-  Big Data
-  Streaming
-  Data Science
-  Data Integration
-  Events
-  Streaming
-  Notifications
-  Monitoring
-  Resource Manager
-  IAM
-  Vault

AWS vs Azure

S. No	Service	Amazon Web Service	Microsoft Azure
1	Compute	Elastic Compute Cloud (EC2)	Azure Virtual Machine (VM)
2	Networking	Virtual Private Cloud (VPC)	Azure Virtual Network (vNet)
3	Storage	Simple Storage Service (S3 bucket)	Azure Storage Account
4	Users and Groups	Identity and Access Management (IAM)	Microsoft Entra ID (Azure Active Directory (AAD)).

Mandatory requirements:

- Create a cloud account.
 - o AWS → Phone number, Email ID, Credit or debit card (Rs 2) [RUPAY card will not be accepted].
 - o Azure → Phone number, Email ID, Credit card (Rs 2).
- You need a subscription for Azure account:
 - o Free trial (for 30 days, you will receive \$100).
 - o Pay-as-you-go account
 - o MSDN subscription (for developers, freelancers, for MCT).
 - o Azure student account
 - o Enterprise subscription (agreement between Microsoft and Wipro).
- For AWS, only email ID (root account) and debit/credit card are required.

Ques – difference between Pricing calculator and TCO calculator.

Azure terms:

- ✓ [Microsoft Azure pricing calculator](#) – this gives the estimated amount for using the resources within a specific region for specific duration.
- ✓ [Microsoft Azure TCO calculator](#) – it's a calculator that shows the total amount that you can save after migrating to Azure cloud from on-premises. TCO = Total Cost of Ownership.
- ✓ [Azure updates](#) – it is a website that provides all the updates on Azure.
- ✓ [Azure status](#) – this website gives the updates on all Azure-based services.
- ✓ [Azure Official documentation](#) – Azure's official site to work with any azure resource.

To access the Azure (ARM) portal, URL: <https://www.portal.azure.com>

Steps for Azure:

1. Microsoft Entra ID
 - ✓ [How to create a user in Azure?](#)
 - ✓ [How to invite a user in Azure?](#)
 - ✓ [How to create users in bulk?](#)
 - ✓ How to give them access to Azure and its resources?
2. Azure Virtual Network (vNet)
 - ✓ How to [create a vNet](#) using address space (192.168.0.0/16).
 - ✓ How to [create 2 subnets](#)
 - i. Subnet-1: 192.168.1.0/24
 - ii. Subnet-2: 192.168.2.0/24
3. Azure Storage Account
 - ✓ How to create a storage account?
 - ✓ How to create
 - i. Azure BLOB storage
 - ii. Azure file share
4. Azure Virtual Machine
 - ✓ Create a windows virtual machine
 - ✓ Create a linux virtual machine

Types of access on subscription:

- Owner
 - If any user gets the owner role, he/she is allowed to create, modify or delete anything within Azure.
- Contributor
 - If any user gets the contributor role, he/she is allowed to create, modify (but no delete) anything within Azure.
- Reader
 - If any user gets the reader role, he/she is allowed only to read (no create, no modify, no delete) anything within Azure.

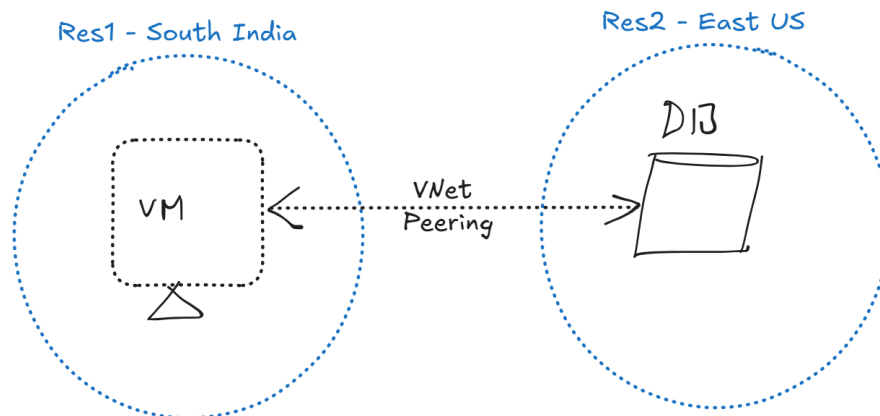
To create users in bulk in Azure:

Pre-requisites are:

- Names of all the users for which the account is getting created.
- Download the template (in CSV format).
- Edit the template.
- Upload this template.

Azure Virtual Network (vNet):

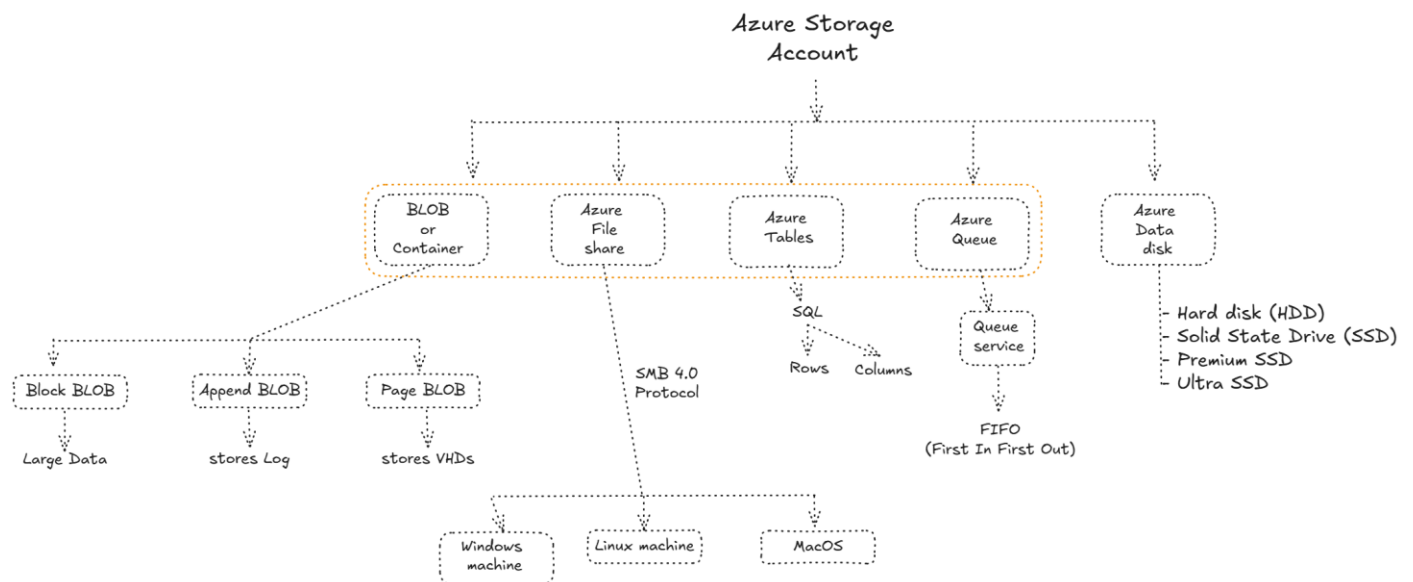
- ✓ vNet is equivalent to on-premises network.
- ✓ It provides the full control over IP address, routing and security,
- ✓ It supports both IPv4 and IPv6.
- ✓ We can create subnets within the network and each subnet can have NSG associated with.
 - NSG
 - Network security Group
 - Its like a firewall for the subnet and its resources.
 - It will only ALLOW the ports that are added to the NSG.
- ✓ vNet uses Vnet peering to access the resources of another vNet in another region.



- ✓ Vnet can be integrated with:
 - Azure Virtual Machines
 - Azure Kubernetes service (AKS)
 - App Gateway
 - Load balancer
 - Firewalls
 - Storage
 - Azure Bastion (secure RDP/SSH)
 - Etc...

Azure Storage Account:

- ✓ Azure storage account is a cloud-based storage service.
- ✓ It offers different types of storages:
 - *BLOB Storage*
 - Binary Large Object
 - This is used for storing unstructured/binary data (images, videos, MP3...)
 - *File Share Storage*
 - Like Network File Share.
 - It uses SMB protocol for sharing the data between different operating system (linux, windows or mac).
 - *Table Storage*
 - For NoSQL, it stores data in key-value format.
 - *Queue Storage*
 - For messaging between components.
 - It works under 2-tier and 3-tier (n-tier) architectures.
 - *Disk storage*
 - Create virtual HDDs/SSDs for Azure virtual machine.



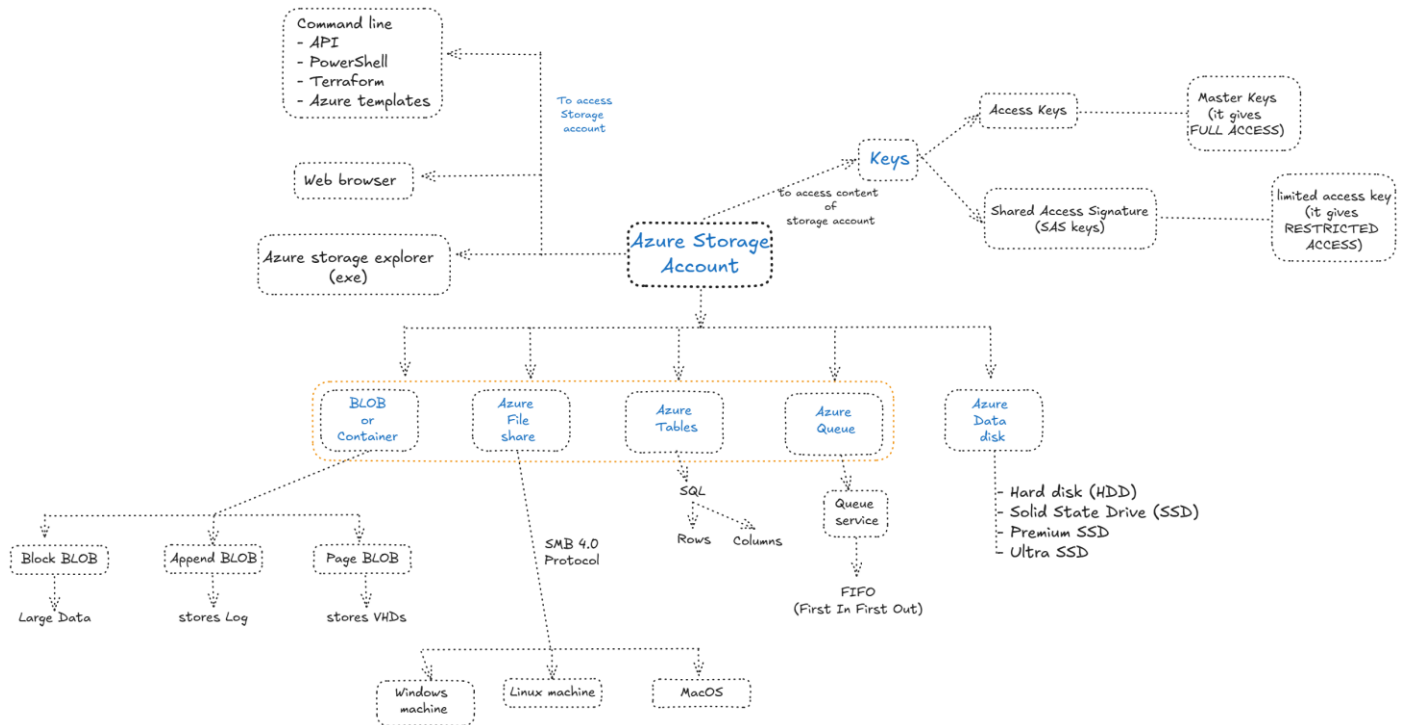
- ✓ Different types of Azure storage account:
 - General-purpose v1 – legacy applications
 - General-purpose v2 – supports all the various features.
 - Blob storage – only for blob.
- ✓ Azure storage account offers redundancy options:
 - *LRS*
 - Locally redundant storage
 - It allows you to store all 3 copies of your data in one data center only.
 - It's the cheapest option to store your data.
 - *ZRS*
 - Zone redundant storage
 - It allows you to store all 3 copies of your data across different zones in different data center.
 - It is costlier than LRS but cheaper than GRS.
 - *GRS*
 - Geo redundant storage
 - It allows you to store all copies of your data in different geographical location (another continent).
 - It's the costliest option for storing the data.

✓ Performance tier:

- Standard – backed by HDD
- Premium – backed by SSD

✓ BLOB Storage tiers

- Hot – for frequently accessed data.
- Cool – for infrequently accessed data (cheaper than Hot tier).
- Archive – for rarely accessed data (very cheap, takes high retrieval time).



Azure Virtual Machine (VM):

- ✓ It's a virtualized instance of a physical server machine.
- ✓ It is just like traditional on-prem server.
- ✓ We can run:
 - Windows server and client OS
 - Linux server & OS.
 - MacOS.
- ✓ Use case:
 - Host public & private websites
 - Host databases
 - Perform development, testing and staging environments.
 - You can also run fully-functional production environment too.
 - Perform migration from on-prem to Azure cloud.
- ✓ Core components:
 - Compute (CPU & RAM)
 - Storage (OS disk, Temporary disk, data disk)
 - Networking (NIC, Public IP, NSG)
- ✓ VM Sizes:
 - B-series → low-cost applications
 - D-series → general purpose
 - E-series → memory optimization
 - F-series → compute optimization
 - L-series → storage optimization
 - N-series → GPU workload
 - H-series → High-performance computing
- ✓ Storage that can be added to a VM:
 - OS disk – the disk that holds/stores OS in it.
 - Temporary disk – acts as a SWAP space for the OS.
 - Data disk – the disk that stores the data in it.

Amazon Web Services (AWS)

- ✓ AWS is a cloud computing platform.
- ✓ Core services offered by AWS are:
 - IAAS → EC2, EBS, VPC
 - PAAS → Elastic Beanstalk, AWS Lambda service
 - SAAS → Amazon Work Docs, Amazon Chime.
- ✓ Compute service
 - EC2
 - Elastic Compute Cloud
 - It's the VM on AWS
 - Lambda function
 - Serverless computing (run the code without servers)
 - ECS/EKS
 - Elastic Container Service / Elastic Kubernetes service
 - Related to container.
 - Elastic Beanstalk
 - PAAS for deploying web applications
- ✓ Storage services
 - S3 bucket
 - Simple Storage Service
 - Stores objects in it.
 - EBS
 - Elastic Block store
 - Used for storing data permanently (Persistent disk)
 - EFS
 - Elastic File system
 - NFS for linux
 - Glacier
 - Long-term archive
 - FSx
 - Managed file system for windows servers & lustre
- ✓ Network service
 - VPC
 - Virtual private cloud
 - Isolated cloud network
 - Route 53
 - DNS service
 - CloudFront
 - Content Delivery Network
 - API Gateway
 - Manages the APIs securely

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Types of users in AWS

1. Root user
 - a. Root user (email ID) is the one who created this AWS account.
2. IAM user
 - a. IAM are the users that gets created within AWS account.
 - b. This helps in providing the access to one or more users to perform a job.

AWS CLI – it's the commands that can be executed on AWS cloud platform to create, modify or delete the resources.

Steps to create VPC in AWS:

- ✓ Create a VPC
- ✓ Create a subnet(s)
- ✓ Create route table
- ✓ Create internet gateway (IGW)
- ✓ Connect internet gateway with route table.
- ✓ Associate route table with subnet(s).
- ✓ Create EC2 instance